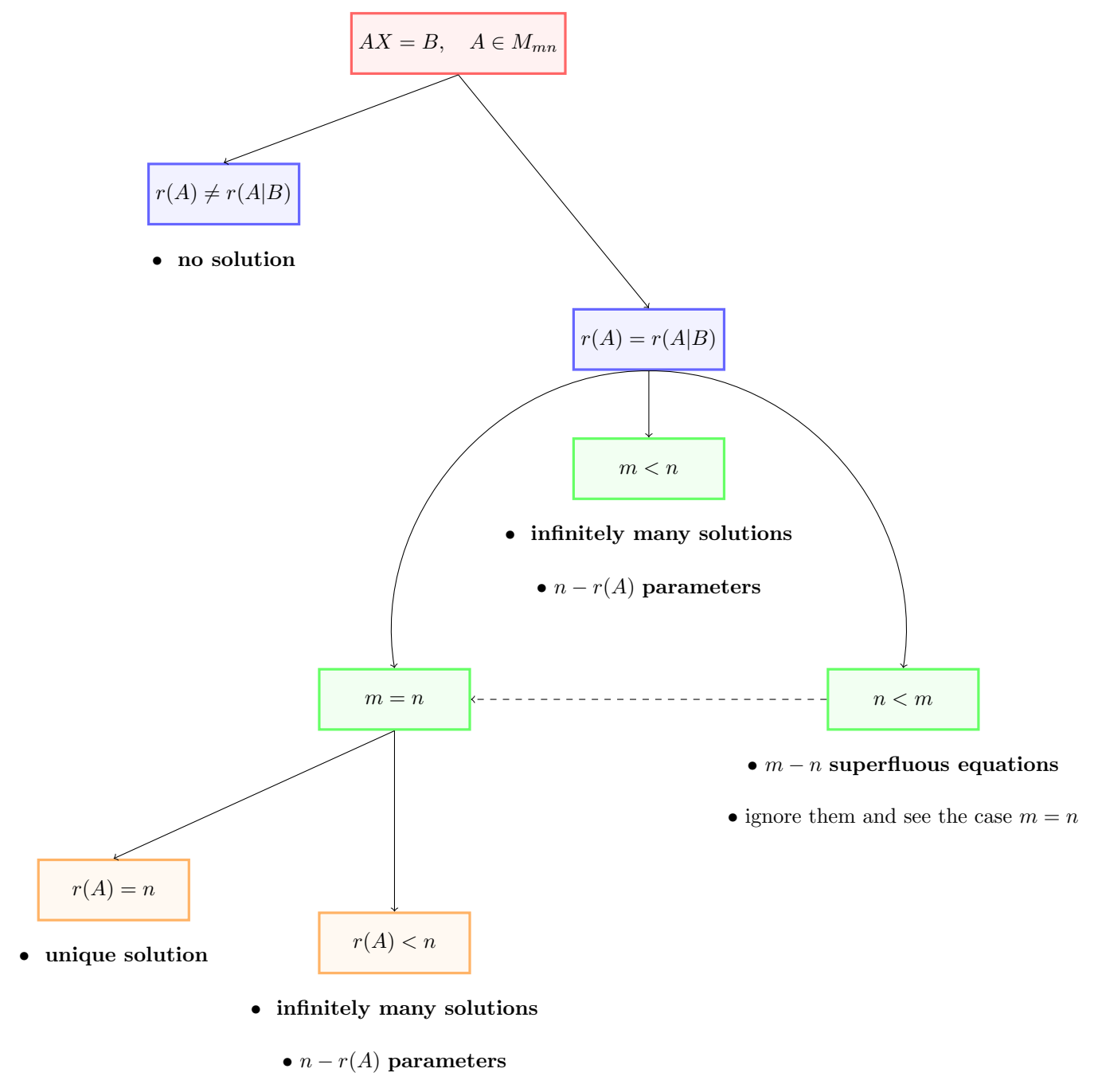

$AX = B, \quad A \in M_{mn}$
$r(A) \neq r(A|B)$
• **no solution**
$r(A) = r(A|B)$
$m < n$
• **infinitely many solutions**
• $n - r(A)$ **parameters**
$m = n$
$n < m$
• $m - n$ **superfluous equations**
• ignore them and see the case $m = n$
$r(A) = n$
• **unique solution**
$r(A) < n$
• **infinitely many solutions**
• $n - r(A)$ **parameters**